Unhang, Fold, and Stack Five T-Shirts SOP (1x Episode: 5 T-Shirts)

One episode takes five T-shirts off their hangers on the garment rack, folds each one on the fold area, and stacks the five folded T-shirts in one stack. The hangers stay on the rail.

Each T-shirt comes off its hanger in two moves. The left gripper lifts the left shoulder off the left end of the hanger. The right gripper then draws the T-shirt down and to the right until the collar slides off the right end. The left gripper and the right gripper carry the T-shirt by its shoulder points to the fold area and lay it down flat, face down, collar toward the front edge.

The fold is done flat on the table in three folds. The left gripper folds the left side in to the **fold line**, bottom corner first and sleeve second. The right gripper does the same on the right side. The left gripper and the right gripper then lift the shoulders and lay them back onto the hem. The left gripper (Config L) or the right gripper (Config R) carries the folded T-shirt to the stack.

The order never changes: unhang, lay flat, fold the left side, fold the right side, fold in half, stack. One T-shirt is off the rail at a time, and it is on the stack before the next one comes off its hanger. The rack never moves and no hanger leaves the rail.

The table is set up in one of two ways. Only the stack spot moves; the rack, the T-shirts on the rail, and the fold area are in the same place in both.

- **Config L:** the stack spot is in the back-left corner, left of the rack's left upright.
- **Config R:** the stack spot is in the back-right corner, right of the rack's right upright.

Where a step depends on the setup it says so on an **IF** line: look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the stack spot is at the back-left (**Config L**) or the back-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Stack-side rule:** in Config L the left gripper carries every folded T-shirt to the stack and the right gripper waits clear of the stack. In Config R the right gripper carries every folded T-shirt to the stack and the left gripper waits clear of the stack. No arm reaches across the table to the stack.
- **Fixed roles:** everything else is the same in both configs. The left gripper lifts the left shoulder off the hanger and folds the left side. The right gripper draws the T-shirt off the hanger and folds the right side. The left gripper and the right gripper carry the T-shirt to the fold area together, lay it down, and fold it in half.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the garment rack across the back with all five T-shirts, the fold area at the center of the table, and the stack spot for this episode's config.
3. The rail is in frame end to end, from its **left end** to its **right end**, so every hanger and every collar can be seen.
4. The whole fold area is in frame, with room around it for both sleeves of a T-shirt laid flat.
5. The left arm and the right arm are at home, the left gripper and the right gripper open.
6. The tabletop holds nothing except the garment rack and robot hardware. The fold area and the stack spot are bare.
7. Nothing blocks the paths between the rail, the fold area, and the stack spot.
8. The left arm reaches both shoulder points of every hanging T-shirt, the whole fold area, and the back-left stack spot. The right arm reaches both shoulder points of every hanging T-shirt, the whole fold area, and the back-right stack spot. Neither arm leans out or stretches to a joint limit to get there.

Materials checklist

1. One **garment rack** stands across the back of the table, in the middle of the table's back edge. Its **left upright** stands at the back left of center and its **right upright** at the back right of center, so the **rail** runs left to right across the back.
2. The rack stands firm on its own base. It does not rock, slide, or lift when a T-shirt is drawn off a hanger, and no gripper ever holds it up.
3. Five **hangers** hang on the rail in **one run**, in the middle of the rail, hook openings facing the back. All five are the same shape and size, with a hook that is not bent.
4. One **T-shirt** hangs on each hanger, front facing the front edge, both shoulders on their hanger ends, the hook standing up out of the collar. T-shirts may touch each other on the rail.
5. The rail is high enough that every hanging T-shirt hangs clear of the tabletop and of a T-shirt lying flat on the fold area.

6. All five T-shirts are short-sleeved, about the same size, dry, clean, and pressed, with no tears and no open seams.
7. Every T-shirt's **collar** opens wide enough to slide off the end of its hanger without pulling out of shape. A T-shirt whose collar will not do that is not staged.
8. The **fold area** at the center of the table is bare and big enough for one T-shirt laid flat with both sleeves out.
9. The **stack spot** for this episode's config is bare:
    - **Config L:** back-left corner, left of the rack's left upright
    - **Config R:** back-right corner, right of the rack's right upright

Workspace layout

- **Garment rack:** across the back of the table, in the middle of the back edge, with the five T-shirts hanging on the **rail**. The rack stays where it is for the whole episode and no gripper ever lifts it.
- **Fold area:** the center of the table, in front of the rack. Each T-shirt is laid flat and folded here, one at a time. The left arm and the right arm reach all of it.
- **Stack spot:** the back-left corner (**Config L**) or the back-right corner (**Config R**), where the five folded T-shirts go in one stack. Only the left gripper reaches the back-left corner and only the right gripper reaches the back-right corner.

The rack and the fold area stay the same for every episode; only the stack spot moves with the config.

Arm assignments

- **Left gripper:** lifts the left shoulder of each T-shirt off the hanger, holds the left shoulder point for the carry and the lay-down, and folds the left side in. In Config L it also carries each folded T-shirt to the stack.
- **Right gripper:** draws each T-shirt off the right end of the hanger, holds the right shoulder point for the carry and the lay-down, and folds the right side in. In Config R it also carries each folded T-shirt to the stack.
- **Left gripper and right gripper together:** carry the T-shirt to the fold area, lay it down, and fold it in half, the left gripper on the left shoulder and the right gripper on the right shoulder.
- While the left gripper folds, the right gripper waits clear of it, and the other way round. The left gripper and the right gripper never crowd the same spot.

Vocabulary

- **T-shirt:** one short-sleeved T-shirt. There are five in an episode and they are worked one at a time.

- **Left and right:** always as seen from the front edge, for the T-shirt, the hanger, and the rail. The **left shoulder** of a hanging T-shirt is the one on the left as seen from the front edge.
- **Next T-shirt:** the T-shirt at the right end of the run on the rail. The first one of the episode is the rightmost of the five; each one after it is the rightmost one left.
- **Hanger:** the frame the T-shirt hangs on. It has a **hook** on top and two **shoulder ends**. The **left end** and **right end** of the hanger are its two shoulder ends.
- **Collar:** the band around the neck opening.
- **Shoulder point:** the top corner of the T-shirt where the shoulder seam meets the sleeve. There is a **left shoulder point** and a **right shoulder point**. The left gripper holds the left shoulder point and the right gripper the right shoulder point from the hanger to the fold area.
- **Held T-shirt:** a T-shirt hanging from the left gripper at the left shoulder point and the right gripper at the right shoulder point, front facing the front edge, the shoulders spread out flat between the left gripper and the right gripper, and the hem hanging free.
- **Face down:** the T-shirt lies with its front on the table and its back showing.
- **Laid flat:** the T-shirt lies face down on the fold area, collar toward the front edge, hem toward the back, both sleeves out to the sides, with no corner, sleeve, or hem folded under.
- **Bottom corner:** one of the two corners where the hem meets a side of the T-shirt. There is a **left bottom corner** and a **right bottom corner**.
- **Sleeve end:** the outer end of a sleeve, at its opening.
- **Fold line:** a straight line from one end of the collar straight back to the hem. There is a **left fold line** and a **right fold line**. The left side is folded in on the left fold line and the right side on the right fold line.
- **Strip:** the T-shirt once both sides are folded in: a long flat piece, the width of the collar, running from the shoulders at the front to the hem at the back, with both sleeves lying inside.
- **Folded T-shirt:** the strip folded in half, shoulders laid on the hem. The **fold edge** is the edge at the front where the strip bends over. The **open edge** is the back edge, where the shoulders and the hem lie on each other.
- **Stack:** the folded T-shirts in one pile on the stack spot, each laid square on the one below it, fold edges toward the front edge.
- **Square on the stack:** a folded T-shirt sits right on top of the one below it, its edges in line with that one's edges on all four sides, nothing hanging over on one side.
- **Lift away:** raise the left gripper or the right gripper, or the item it holds, straight up until there is daylight under it, before anything moves sideways.
- **Carry level:** move the item across at one height, without turning, swinging, or shaking it.
- **Settled:** the item stays put when the left gripper or the right gripper lifts away, with nothing sliding, springing open, or falling.

- **Let-go point:** the left gripper and the right gripper open only at one of these: the left gripper on the fold area once a left corner or the left sleeve lies where it was laid; the right gripper on the fold area once a right corner or the right sleeve lies where it was laid; the left gripper and the right gripper together once the T-shirt is laid flat or the shoulders lie on the hem; the left gripper (Config L) or the right gripper (Config R) once the folded T-shirt lies on the stack spot or square on the stack.

Steps

Run Steps 1 to 6 five times, once per T-shirt. Then run Step 7 once. One T-shirt is off the rail at a time, and it is on the stack before the next one comes off its hanger.

Only Step 6 depends on the config: in Config L the left gripper carries the folded T-shirt to the stack, and in Config R the right gripper does. Steps 1 to 5 are the same in both configs.

Any grasp or fold may be retried. If the left gripper or the right gripper slips off, or a corner it laid lands in the wrong place, that same gripper opens, takes the same point again, and lays it again. A retry is always made by the gripper the step names. Nothing is ever forced to save a retry.

Step 1: Take the next T-shirt off its hanger

Goal: the next T-shirt is a held T-shirt, off its hanger, and the empty hanger still hangs on the rail.

1.1 Lift the left shoulder off

- The **left gripper** pinches the **left shoulder point** of the **next T-shirt**, on the cloth just past the left end of the hanger.
- The **left gripper** lifts the left shoulder up and off the left end of the hanger, then brings it a little in and down, and keeps hold.
- The **right gripper** waits clear of the rail.

1.2 Draw the T-shirt off the right end

- The **right gripper** pinches the **right shoulder point**.
- The **right gripper** draws the T-shirt down and to the right, so the collar slides along the right side of the hanger and off the right end.
- The **left gripper** keeps hold of the left shoulder point and moves with the T-shirt, so the shoulders stay spread between the two grippers.

Take only the next T-shirt. Never pull a T-shirt straight down off its hanger, never take the hanger off the rail, and never lift a T-shirt by its collar, a sleeve, or the hem. If the collar catches on the hanger, the **right gripper** brings the T-shirt back up a little and draws it off again.

Check: the T-shirt is a **held T-shirt**: the **left gripper** on the left shoulder point, the **right gripper** on the right shoulder point, front facing the front edge, shoulders spread between them, hem hanging free. The empty hanger still hangs on the rail with its hook over the rail. The other T-shirts still hang on their hangers.

Step 2: Lay the T-shirt flat on the fold area

Goal: the T-shirt is **laid flat** on the fold area, face down, collar toward the front edge.

- The **left gripper** and the **right gripper** carry the T-shirt level, side by side, forward to the fold area.
- The **left gripper** and the **right gripper** bring it down together until the hem touches the back of the fold area.
- The **left gripper** and the **right gripper** keep moving toward the front edge as they come down, so the T-shirt lays down onto its front, hem first and collar last.
- With the collar on the table toward the front edge, the **left gripper** and the **right gripper** open and lift away.

Check: the T-shirt is laid flat: face down, collar toward the front edge, hem toward the back, both sleeves out to the sides, no corner, sleeve, or hem folded under, and no fold across the body.

If a left corner or the left sleeve is folded under, the **left gripper** pinches it and pulls it out flat. If a right corner or the right sleeve is folded under, the **right gripper** pinches it and pulls it out flat. If the T-shirt is face up, twisted, or bunched, the **left gripper** and the **right gripper** take the two shoulder points again, lift the T-shirt off the table until the hem hangs free, and lay it down again.

Step 3: Fold the left side in

Goal: the left side lies folded in on the **left fold line**, sleeve on top, with a straight crease.

3.1 Fold in the left bottom corner

- The **left gripper** pinches the **left bottom corner**, lifts it, and carries it to the right over the T-shirt.
- The **left gripper** lays it down so the left side bends over along the left fold line, the crease running straight from the left end of the collar to the hem.
- The **left gripper** opens and lifts away.

3.2 Fold in the left sleeve

- The **left gripper** pinches the **left sleeve end**, lifts it, and carries it to the right over the folded side.
- The **left gripper** lays the sleeve flat on top of the folded side, inside the left fold line.
- The **left gripper** opens and lifts away.

The **right gripper** waits clear of the fold area the whole step.

Check: the left side lies folded on the left fold line with one straight crease from the collar to the hem. The sleeve lies flat on top, inside the crease, nothing sticking out past the crease.

If the crease runs crooked or does not meet the collar, the **left gripper** lifts the left bottom corner and lays it again. If the sleeve sticks out past the crease or is folded under, the **left gripper** lifts the sleeve end and lays it again.

Step 4: Fold the right side in

Goal: the T-shirt is a **strip**: both sides folded in on their fold lines, sleeves on top.

4.1 Fold in the right bottom corner

- The **right gripper** pinches the **right bottom corner**, lifts it, and carries it to the left over the T-shirt.
- The **right gripper** lays it down so the right side bends over along the right fold line, the crease running straight from the right end of the collar to the hem.
- The **right gripper** opens and lifts away.

4.2 Fold in the right sleeve

- The **right gripper** pinches the **right sleeve end**, lifts it, and carries it to the left over the folded side.
- The **right gripper** lays the sleeve flat on top of the folded side, inside the right fold line.
- The **right gripper** opens and lifts away.

The **left gripper** waits clear of the fold area the whole step.

Check: the T-shirt is a strip the width of the collar, with two straight creases running side by side from the collar to the hem. Both sleeves lie flat inside the creases. The left side has not opened.

If the right crease runs crooked, the **right gripper** lifts the right bottom corner and lays it again. If the right sleeve sticks out, the **right gripper** lifts the right sleeve end and lays it again. If the left sleeve sticks out, the **left gripper** lifts the left sleeve end and lays it again. If the left side opened, the **left gripper** runs Step 3 again.

Step 5: Fold the T-shirt in half

Goal: a **folded T-shirt**: the shoulders laid on the hem, fold edge toward the front edge.

- The **left gripper** pinches the strip at its left shoulder corner and the **right gripper** at its right shoulder corner, at the front end of the strip.
- The **left gripper** and the **right gripper** lift the shoulders together and carry them level toward the back, over the top of the strip.

- The **left gripper** lays the left shoulder corner on the left bottom corner, and the **right gripper** lays the right shoulder corner on the right bottom corner.
- The **left gripper** and the **right gripper** open and lift away.

Check: the folded T-shirt lies flat, the shoulder edge on top of the hem edge, corners on corners, the fold edge straight across the front. No sleeve or corner sticks out from between the layers.

If the shoulders landed off the hem or crooked, the **left gripper** and the **right gripper** take the shoulder corners again, lift them a little, and lay them again on the hem.

Step 6: Put the folded T-shirt on the stack

Goal: the folded T-shirt lies **square on the stack**, fold edge toward the front edge.

- **IF the stack spot is at the back-left (Config L):** the **left gripper** pinches the folded T-shirt at the middle of its fold edge, lifts it a little, and carries it level to the left, to the stack spot. The **right gripper** waits clear of the stack.
- **IF the stack spot is at the back-right (Config R):** the **right gripper** pinches the folded T-shirt at the middle of its fold edge, lifts it a little, and carries it level to the right, to the stack spot. The **left gripper** waits clear of the stack.
- **IF Config L:** the **left gripper** brings the folded T-shirt down, open edge first, and lays it flat as it comes down, fold edge toward the front edge. The first one goes in the middle of the bare stack spot and each one after it square on top of the stack. Then the **left gripper** opens and lifts away.
- **IF Config R:** the **right gripper** brings the folded T-shirt down, open edge first, and lays it flat as it comes down, fold edge toward the front edge. The first one goes in the middle of the bare stack spot and each one after it square on top of the stack. Then the **right gripper** opens and lifts away.
- Go back to Step 1 for the next T-shirt. After the fifth T-shirt, go on to Step 7.

Carry the folded T-shirt low and keep it folded. Never drag it across the table and never drop it onto the stack.

Check: the folded T-shirt lies square on the stack, fold edge toward the front edge, still folded, and **settled**. The stack below it has not moved.

If it landed off square, the **left gripper** (Config L) or the **right gripper** (Config R) lifts it a little by the middle of the fold edge and lays it again. Neither the left gripper nor the right gripper ever pushes the whole stack.

Step 7: End the episode

Goal: recording ends with five folded T-shirts in one stack and five empty hangers on the rail.

- Look once across the table: five folded T-shirts square in one stack, fold edges toward the front, five empty hangers on the rail, and the fold area bare.
- Return the left arm and the right arm home, the **left gripper** and the **right gripper** open. Homing is the last thing the arms do.
- Stop recording.

Check: the left arm and the right arm are at home, the **left gripper** and the **right gripper** are open, and neither holds anything.

Expected state: the stack and the empty hangers are still, the rack has not moved, and the recording has stopped.

After the episode: reset the workspace

This reset is not recorded.

1. Unfold each T-shirt from the stack and smooth it out. Press it if it is creased from the fold.
2. Hang each T-shirt back on a hanger on the rail, through the collar, both shoulders on their hanger ends, front facing the front edge.
3. Leave the five hangers in one run in the middle of the rail, hook openings facing the back.
4. Check every collar still slides off the end of its hanger without pulling out of shape, and that no collar is stretched or wavy.
5. Check all five hangers: no bent hook, no cracked shoulder end.
6. Check the rack stands firm, the rail is straight, and the uprights have not shifted.
7. Replace anything damaged: a T-shirt that is torn, stained, or damp; a T-shirt whose collar is stretched; a bent hanger; or a rack that rocks.
8. Clear and wipe the fold area. Make sure the stack spot for the next episode's config is bare: back-left corner (Config L) or back-right corner (Config R).
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain several violations; tag each one separately. Keep the episode with its violation tags. Do not delete it just because a

rule was broken.

Tag a violation only when something is forced, the order is broken, or the episode moves on with a fault left in place.

Violations

Violation: Config misaligned

- **Visible cue:** what the arms do does not match the config on the table: the stack is built somewhere other than the stack spot for the config; the right gripper carries a folded T-shirt to the stack in Config L, or the left gripper in Config R; the left gripper or the right gripper reaches across the table to the stack; or the wrong IF line is followed.
- **SOP rule broken:** Step 6, the stack-side rule (the left gripper in Config L and the right gripper in Config R carry every folded T-shirt to the stack, no arm reaches across the table to the stack, and the IF line followed is the one for the config on the table).
- **Coaching note:** look where the stack spot is before the first carry, then follow that config's IF line in Step 6.

Violation: Wrong order, or two T-shirts worked at once

- **Visible cue:** a fold starts before the T-shirt is laid flat; the right side is folded before the left; the T-shirt is folded in half before both sides are in; or a second T-shirt comes off the rail before the last one is on the stack.
- **SOP rule broken:** Steps 1 to 6, unhang, lay flat, fold the left side, fold the right side, fold in half, then stack, in that order, one T-shirt at a time.
- **Coaching note:** finish the T-shirt on the stack before you touch the next one.

Violation: T-shirt taken out of order on the rail

- **Visible cue:** a T-shirt other than the rightmost one left on the rail is taken off.
- **SOP rule broken:** Step 1, the next T-shirt is always the one at the right end of the run.
- **Coaching note:** always the rightmost one.

Violation: T-shirt pulled off the hanger the wrong way

- **Visible cue:** the T-shirt is pulled straight down off the hanger, the right gripper draws it off before the left shoulder is off the left end, the collar is dragged off the left end, or the T-shirt is lifted by its collar, a sleeve, or the hem.
- **SOP rule broken:** Steps 1.1 and 1.2, the left gripper lifts the left shoulder off the left end first, then the right gripper draws the T-shirt down and to the right until the collar slides off the right end, both holding the shoulder points.
- **Coaching note:** left shoulder up and off, then draw it off to the right.

Violation: Collar stretched or T-shirt forced off

- **Visible cue:** the collar pulls out of shape or a seam goes tight as it comes off the hanger, or the T-shirt is yanked when the collar catches instead of being brought back up and drawn off again.
- **SOP rule broken:** Step 1.2, the collar slides off the right end without force; if it catches, the right gripper brings the T-shirt back up a little and draws it off again.
- **Coaching note:** if it catches, go back up and draw again. Never yank.

Violation: Hanger taken off or knocked off the rail

- **Visible cue:** a hanger lifts off the rail, comes away with the T-shirt, falls, or hangs on another hanger's hook, and the episode moves on; or the left gripper or the right gripper takes a hanger by its hook.
- **SOP rule broken:** Steps 1.1 and 1.2, the T-shirt comes off its hanger and the hanger stays on the rail with its hook over the rail.
- **Coaching note:** the hanger stays on the rail. Only the T-shirt moves.

Violation: T-shirt not held by both shoulder points for the carry

- **Visible cue:** the T-shirt is carried to the fold area by the left gripper alone or the right gripper alone, by a sleeve, the collar, or the hem, or it drags on the table or the rack on the way.
- **SOP rule broken:** Steps 1.2 and 2, the left gripper holds the left shoulder point and the right gripper the right shoulder point from the hanger to the fold area, and the T-shirt is carried level.
- **Coaching note:** left gripper on the left shoulder point, right gripper on the right, carried level.

Violation: T-shirt laid down the wrong way

- **Visible cue:** the T-shirt is laid face up, collar toward the back or a side, dropped onto the table, or laid outside the fold area.
- **SOP rule broken:** Step 2, the T-shirt is laid face down on the fold area, hem first and collar last, collar toward the front edge.
- **Coaching note:** hem down at the back, then move toward the front as it goes down.

Violation: Fold started on a T-shirt not laid flat

- **Visible cue:** a side is folded while a corner, a sleeve, or the hem is folded under, while the T-shirt is twisted or bunched, or while there is a fold across the body.
- **SOP rule broken:** Step 2, the T-shirt is laid flat, with nothing folded under, before any fold starts.
- **Coaching note:** look at all four corners and both sleeves before you fold. Pull out anything under.

Violation: Side folded the wrong way

- **Visible cue:** a sleeve is folded in before its bottom corner; the right gripper folds the left side or the left gripper folds the right side; a side is folded from the middle of the body or by the hem instead of the bottom corner and the sleeve end; or both sides are folded at once.
- **SOP rule broken:** Steps 3 and 4, the left gripper folds the left side and the right gripper the right side, bottom corner first and sleeve second, left side then right side.
- **Coaching note:** bottom corner, then sleeve. Left gripper on the left side, right gripper on the right side.

Violation: Crease crooked or off the fold line

- **Visible cue:** a side crease runs crooked, does not meet the end of the collar, or makes the strip wider at one end, and the episode moves on to the next fold.
- **SOP rule broken:** Steps 3.1 and 4.1, each side bends over on its fold line, the crease running straight from the end of the collar to the hem; a crooked crease is laid again before moving on.
- **Coaching note:** line the crease up with the end of the collar. If it is crooked, lay the corner again.

Violation: Sleeve left sticking out

- **Visible cue:** a sleeve sticks out past its crease, is folded under, or is left unfolded, and the episode moves on.
- **SOP rule broken:** Steps 3.2 and 4.2, each sleeve lies flat on top of its folded side, inside the crease.
- **Coaching note:** lay the sleeve inside the crease. If it sticks out, lay it again.

Violation: Half fold done the wrong way

- **Visible cue:** the hem is folded onto the shoulders instead of the shoulders onto the hem; the strip is folded from the side instead of end to end; the half fold is made by the left gripper alone or the right gripper alone; or the shoulders land off the hem or crooked and the episode moves on.
- **SOP rule broken:** Step 5, the left gripper takes the left shoulder corner and the right gripper the right shoulder corner, they carry them back over the strip, and lay them on the hem, corners on corners.
- **Coaching note:** left gripper on the left shoulder corner, right gripper on the right, back over the top, corners on corners.

Violation: Folded T-shirt carried the wrong way

- **Visible cue:** the folded T-shirt is carried by an open edge or a corner instead of the middle of the fold edge; it is dragged across the table; it is carried high or swung; or it opens up on the way.

- **SOP rule broken:** Step 6, the left gripper (Config L) or the right gripper (Config R) pinches the middle of the fold edge, lifts it a little, and carries it level and low, still folded.
- **Coaching note:** middle of the fold edge, low and level.

Violation: Folded T-shirt dropped onto the stack

- **Visible cue:** the left gripper (Config L) or the right gripper (Config R) opens above the stack so the folded T-shirt falls the last part of the way, lands opened, or slides off; or the left gripper or the right gripper opens anywhere other than at a let-go point.
- **SOP rule broken:** Step 6, the folded T-shirt is brought down and laid flat before the left gripper (Config L) or the right gripper (Config R) opens, and the left gripper and the right gripper open nowhere but at a let-go point.
- **Coaching note:** all the way down, lay it flat, then let go.

Violation: Folded T-shirt not square on the stack

- **Visible cue:** a folded T-shirt hangs over the one below on one side, sits turned, has its fold edge toward the back or a side, or the first one is put off the middle of the stack spot, and the next T-shirt comes off the rail.
- **SOP rule broken:** Step 6, each folded T-shirt lies square on the stack, fold edge toward the front edge, and an off-square one is lifted and laid again.
- **Coaching note:** edges on edges, fold to the front. If it is off, lift it and lay it again.

Violation: Stack pushed or slid

- **Visible cue:** the left gripper or the right gripper pushes, slides, or squares the whole stack, or slides a folded T-shirt along the top of the stack instead of lifting it and laying it again.
- **SOP rule broken:** Step 6, an off-square T-shirt is fixed by lifting it a little by the middle of the fold edge and laying it again; the stack is never pushed.
- **Coaching note:** lift it and lay it again. Nothing gets pushed.

Violation: Rack moved or lifted

- **Visible cue:** the rack rocks, slides, tips, or lifts off the tabletop; the rail bends under a pull; or the left gripper or the right gripper holds, drags, leans on, or turns the rack.
- **SOP rule broken:** Steps 1.1 and 1.2, the rack stands on its own base and never moves, and neither the left gripper nor the right gripper touches it.
- **Coaching note:** the rack stands on its own. If it is moving, stop and set it back.

Violation: Arms swapped roles or crowded the same spot

- **Visible cue:** the right gripper lifts the left shoulder off or folds the left side; the left gripper draws the T-shirt off or folds the right side; the right gripper works over the fold area while the left gripper folds, or the other way round; or the left gripper and the right gripper work the same spot, touch each other, or one has to move before the other can reach.

- **SOP rule broken:** Steps 1 to 5, the left gripper lifts the left shoulder off and folds the left side, the right gripper draws the T-shirt off and folds the right side, the right gripper waits clear while the left gripper folds, and the left gripper waits clear while the right gripper folds.
- **Coaching note:** left does the left, right does the right. Neither one does the other's work.

Violation: Item dropped or something knocked over

- **Visible cue:** a T-shirt falls onto the tabletop or the floor; a hanging T-shirt is knocked off its hanger; the left gripper or the right gripper strikes the rack or an upright; the stack is knocked over or out of place; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, each item stays on a clear path, the T-shirts not yet worked stay on their hangers, and the left gripper and the right gripper open only at a let-go point.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with fewer than five T-shirts in the stack, a T-shirt still on a hanger or on the fold area, a hanger off the rail, or the stack not square; or either arm is away from home, the left gripper or the right gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 7, look once across the table, return the left arm and the right arm home with the left gripper and the right gripper open, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Bad T-shirt:** a torn T-shirt or open seam, or a collar that will not slide off the end of its hanger without pulling out of shape. Replace it before the next episode.
- **Bad hanger:** a bent or broken hook or shoulder end. Replace it before the next episode.
- **Rack fault:** the rack will not stand straight, an upright works loose, or the rail bends. Repair or replace it before the next episode.
- **Rail at the wrong height:** a hanging T-shirt touches the tabletop or a T-shirt lying on the fold area.

Annotation subtasks (from SOP)

1. Lift the left shoulder off the hanger

2. Draw the T-shirt off the right end of the hanger
3. Lay the T-shirt flat on the fold area
4. Fold in the left bottom corner
5. Fold in the left sleeve
6. Fold in the right bottom corner
7. Fold in the right sleeve
8. Fold the shoulders back onto the hem
9. Put the folded T-shirt on the stack
10. Return the left arm and the right arm home and end the episode